

Technology Stack

Date 30 September 2026

Team ID SWTID-2026-5550

Project Name Legalease: AI-powered legal document generator

Maximum Marks 2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js, HTML5, CSS3, JavaScript	React.js provides a reusable component-based interface for building a responsive and user-friendly legal document generation dashboard.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, FastAPI	FastAPI provides a lightweight, high-performance backend for document generation, validation, authentication, and AI service integration.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL	PostgreSQL provides reliable relational storage for users, document metadata, templates, generated documents, and application records.

Technology Stack Details (Continued)

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Docker, AWS	Docker packages the application consistently for deployment, while AWS provides scalable cloud infrastructure for hosting the Legalease application.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub, GitHub Actions	GitHub enables collaborative version control and source management, while GitHub Actions supports automated testing and deployment workflows.

Technology Stack Summary: Legalease uses a modern web architecture with React.js for the user interface, Python/FastAPI for backend services, PostgreSQL for structured data, Docker/AWS for deployment, and GitHub/GitHub Actions for version control and CI/CD.

Technology Stack Details (Continued)

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	OpenAI API, PyPDF2, python-docx	The OpenAI API enables AI-assisted legal document generation and refinement. PyPDF2 supports PDF document processing, while python-docx supports creation and processing of editable Word documents.